Jatoth Manjunath Naik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
Alphores Junior College	Class XII (Telangana State Board of Intermediate Education)	2019	85.90%
Rao's Techno International School	Class X (Board of Secondary Education Telangana State)	2017	9.00%

TECHNICAL SKILLS

- **Programming & Core:** Python, C, JavaScript, OOP, Git, Problem Solving, SQL
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, RNNs, LSTMs, Vision Transformers, Model Evaluation & Tuning
- **NLP, GenAI & Computer Vision:** Transformers, CLIP, RAG, LangChain, FAISS, LoRA, Llama 3, Sentence-BERT, DeBERTa, OpenCV
- **Data Science & Development:** NumPy, Pandas, SciPy, Matplotlib, Power BI, FastAPI, Flask, Streamlit, Docker, Linux

PROJECTS

TEST-TIME ADAPTATION FOR VISION-LANGUAGE MODELS

- **Conducting a structured study of Online Test-Time Adaptation**, covering entropy-based adaptation, continual adaptation, prompt tuning, reliable sample selection, prototype-based adaptation, bias mitigation, and open-set OOD detection.
- **Investigating reliability challenges in CLIP-based vision-language models**, including class-wise confidence bias, long-tail degradation, unreliable pseudo-labels, cache contamination, and unknown out-of-distribution samples.
- **Developing a bias- and OOD-aware reliability framework** to separate incoming samples into reliable known, uncertain, and likely OOD categories for safer online adaptation.

MULTIMODAL RAG FOR INTELLIGENT PRODUCT UNDERSTANDING

- **Engineered a multimodal AI system** integrating **CLIP ViT-B/16** vision embeddings and **DeBERTa-v3** text encoders with cross-modal attention, enabling semantic product retrieval and attribute enrichment across **50K+ catalog items**.
- **Developed a RAG-based generative Q&A pipeline** using **LangChain, FAISS, and LoRA-tuned Llama 3**, achieving **15% higher retrieval accuracy** and reducing hallucinations by **>25%** over baseline approaches.
- **Delivered an end-to-end FastAPI + Streamlit prototype** for catalog tagging and duplicate detection, achieving **0.87 attribute F1**, a **13% improvement in nDCG@10**, and **sub-second query latency**.

CAUSAL SENSE -- LLM-DRIVEN CAUSAL REASONING & EXPLANATION ENGINE

- **Built a causal reasoning framework** to extract cause-effect relationships from scientific and industrial text using **LangChain RAG, Sentence-BERT, and LLM-generated triples**, creating a weighted causal knowledge graph with citations.
- **Implemented explainable "why" and "what-if" reasoning** using **NetworkX graph traversal and Llama 3**, achieving **0.79 causal extraction F1** and improving faithfulness by **14%** over baseline LLMs.
- **Developed an interactive FastAPI + Streamlit system** for graph visualization and counterfactual simulation, achieving **15% higher accuracy** and a **4.5/5 user-rated clarity score** for interpretable reasoning.

MULTI-LABEL EMOTION RECOGNITION WITH XLM-RoBERTa

- **Fine-tuned XLM-RoBERTa** for multilingual multi-label emotion classification on **English and Romanian** data from **SemEval 2025 Task 11**, achieving **0.50-0.53 macro F1**.
- **Designed a class-imbalance-aware training pipeline** using **BCEWithLogitsLoss with pos_weight and AdamW**, improving learning across less-frequent emotion labels.
- **Integrated LIME and BERTViz** to provide **word-level and attention-based explanations**, improving interpretability

of multilingual emotion predictions.

POSITION OF RESPONSIBILITY

Placement Coordinator, IIT Hyderabad

- **Coordinating placement-related communication and outreach** with recruiters and students, while supporting shortlisting, scheduling, and recruitment process execution

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) -- Microsoft

Machine Learning Project Certification -- Face Recognition using OpenCV

Web Development Technologies -- React.js, Node.js, JavaScript, HTML, CSS, SQL, Python

Developer Foundations (Git) -- Version Control & Collaboration

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Probability & Stochastic Processes, Topics in Optimization

HOBBIES & INTERESTS

Badminton, Volleyball, Travelling, and Exploring Emerging Technologies